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Level 2 — 25 marks/question

L2-Q1. (M) Matrix Mult.

j

CODE

```
import java.util.Scanner;
public class Multiplication{
    public static void main(String args[]){
        Scanner sc= new Scanner(System.in);
        System.out.println("Mat1 = ");
        int a[][]=new int[2][2];
        for(int i=0;i<2;i++){
            for(int j=0;j<2;j++){
                a[i][j]=sc.nextInt();
                System.out.print(a[i][j]+"  ");
            }
            System.out.println();
        }
        int b[][]=new int[2][2];
        System.out.println("Mat2 = ");
        for (int j=0;j<2;j++){
            for (int k=0;k<2;k++){
                b[j][k]=sc.nextInt();
                System.out.print(b[j][k]+"  ");
            }
            System.out.println();
        }
        int m[][]=new int[2][2];
        System.out.println("Mat multiplication = ");
        m[0][0]=(a[0][0]*b[0][0])+(a[0][1]*b[1][0]);
        m[0][1]=(a[0][0]*b[0][1])+(a[0][1]*b[1][1]);
        m[1][0]=(a[1][0]*b[0][0])+(a[1][1]*b[1][0]);
        m[1][1]=(a[1][0]*b[0][1])+(a[1][1]*b[1][1]);
        for (int l=0;l<2;l++){
            for (int r=0;r<2;r++){
                System.out.print(m[l][r]+"  ");
            }
            System.out.println();
        }

    }
}
```

INPUT

```
1 2
5 3
2 3
4 1
```

OUTPUT

```
Mat1 =
1  2
5  3
Mat2 =
2  3
4  1
Mat sum =
10  5
22  18
```

L2-Q2. (M) Perfect Num. Gen.

j

CODE

```
import java.util.Scanner;
public class Per{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        boolean flag=true;
        int a=1;
        int b=0;
        while(b!=n){
            int f=0;
            while(flag){
                int i=1;
                if(a%i==0){
                f+=i;
                }
                i+=1;
                a+=1;
                if(b==n){
                    flag=false;
                }
            }
            if(f==n){
                System.out.print(f+" ,");
                b+=1;
            }

        }

    }
}
```

INPUT

```
3
```

OUTPUT

```
Error: Execution timeout (>5 seconds) — possible infinite loop
```

L2-Q3. (M) Tax

j

CODE

```
import java.util.Scanner;
public class Tax{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        int in=sc.nextInt();
        double tax=0;
        if(in<=150000){
           tax=0;
        }
        else if(in>150000 & in<=300000){
            tax=0.10*in;
        }
        else if(in>300000 & in <500000){
            tax=0.20*in;
        }
        else{
            tax=0.30*in;
        }
        System.out.println("Tax= "+tax);
    }
}
```

INPUT

```
200000
```

OUTPUT

```
Tax= 20000.0
```

L2-Q4. (M) Palindrome

j

CODE

```
import java.util.Scanner;
public class Pal{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        System.out.println("Case 1: Given string is palindrome or not");
        System.out.println("Case 2: Given number is palindrome or not");
        int c=sc.nextInt();
        sc.nextLine();
        System.out.println("Case = "+c);
        if(c==1){
            String st=sc.nextLine();
            String str=st.toUpperCase();
            System.out.println("String = "+st);
            String r="";
            for (int i=0;i<str.length();i++){
                r=r+str.charAt(i);
            }
            if(str.equals(r)){
                System.out.println("Palindrome");
            }
            else{
                System.out.println("Not a Palindrome");
            }
        }
        else if(c==2){
            System.out.print("Number = ");
            int n=sc.nextInt();
            System.out.println(n);
            String num=""+n;
            int l=num.length();
            String rev="";
            for(int j=0;j<l;j++){
                rev=rev+num.charAt(j);
            }
            if(num.equals(rev)){
                System.out.println("Palindrome");
            }
            else{
                System.out.println("Not a Palindrome");
            }
        }
        else{
            System.out.println("Invalid Choice");
        }
    }
}
```

INPUT

```
1
MADAM
```

OUTPUT

```
Case 1: Given string is palindrome or not
Case 2: Given number is palindrome or not
Case = 1
String = MADAM
Palindrome
```